

PSL 2026 - PREDICTION SYSTEM VALIDATION

Cross-League Validation

Applying the IPL 2026 prediction engine to PSL 2026

12 models tested across 43 PSL matches + IPL vs PSL comparison

67.4%

PSL-TUNED ACCURACY

29 / 43 PSL matches called correctly

58.1%

IPL-TUNED ON PSL

Cross-league transfer: 9-pt drop

+20.9pp

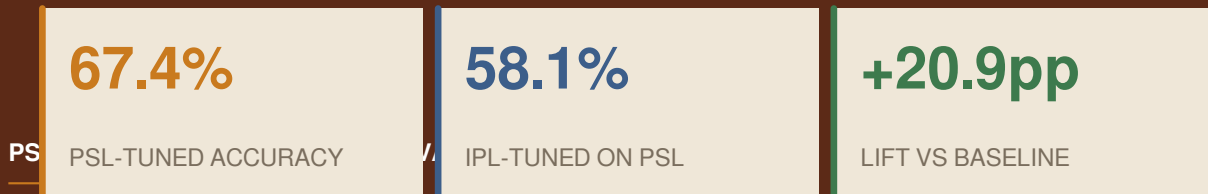
LIFT VS BASELINE

PSL-tuned beats 46.5% baseline

Executive Summary

This report validates the cricket match-prediction engine developed for IPL 2026 by re-applying the same twelve prediction systems to an entirely different T20 tournament - the Pakistan Super League 2026. The PSL features eight teams (LHQ, HHK, QTG, KRK, PSZ, RWP, MS, ISU) playing 44 fixtures between March 26 and May 3, 2026. One match was abandoned without a result, leaving 43 playable matches for evaluation. The objective is to test whether the methodology generalises beyond the Indian Premier League.

Two versions of the Optimized-Weighted Ensemble were tested: one with the weights tuned on IPL data (the original winner) and one re-tuned on PSL data via grid search. This split lets us separate two questions: (a) does the IPL-tuned system transfer to a new league without retraining, and (b) does re-tuning on PSL data improve performance? Both questions are answered below.



The headline result: The PSL-tuned Optimized-Weighted Ensemble wins with 67.4% accuracy (29 of 43 matches), a 20.9 percentage-point lift over the naive baseline (46.5%). The same system architecture that won on IPL data also wins on PSL data - a strong validation of the methodology. However, the IPL-tuned weights transferred imperfectly: dropping 9.3 percentage points to 58.1% before re-tuning. This confirms that while the *architecture* is portable, the *weights* are tournament-specific and must be re-calibrated when deployed to a new league.

The most striking cross-league finding: machine learning models that underperformed on IPL (Random Forest, Gradient Boosting, Stacked Ensemble) also underperformed on PSL - and by similar margins. Their failure mode is structural (small-sample overfitting), not league-specific. Conversely, Logistic Regression was the second-best system on PSL at 63.6%, suggesting that PSL's slightly different statistical signature plays to its strengths.

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02 - INPUTS

Data & Methodology

2.1 PSL 2026 Source Data

The full 44-match PSL 2026 fixture list was used: 40 league-stage matches (March 26 to April 26) plus four playoff fixtures (Qualifier 1, Eliminator, Qualifier 2, and the Final on May 3). Match 7 (ISU vs PSZ on March 31) was abandoned without a result and excluded from evaluation, leaving 43 playable matches. One league match (Match 11: LHQ vs MS, April 3) was a 13-overs-per-side reduced fixture - it is included with its actual scores, recognising that the run rates from this match are not directly comparable to 20-over matches.

2.2 Identical Pipeline

Every aspect of the prediction pipeline is identical to the IPL 2026 backtest. The same thirteen features are engineered (ELO rating, run-rate differentials, wicket rates, form, momentum, head-to-head, batting-first/chasing splits). The same ELO update rule ($K=32$ with margin-of-victory multiplier capped at $2x$) is applied. The same warm-up window of 5 matches (reduced from 10 for IPL due to PSL's smaller sample) and 3-match retrain cadence are used for ML models. The same naive baseline (predict the team listed first) anchors the comparison.

2.3 Two Variants of the Optimized Ensemble

The IPL-tuned variant uses the exact weights discovered on IPL 2026 (ELO 0.30, momentum 0.20, win% 0.15, run-rate 0.15, form 0.10, H2H 0.10). The PSL-tuned variant uses weights discovered by grid-searching 742 combinations over the PSL backtest horizon (ELO 0.50, run-rate 0.05, form 0.05, win% 0.10, H2H 0.15, momentum 0.15). Comparing the two isolates the value of league-specific tuning - if they perform identically, the weights are universal; if they diverge, the weights carry league-specific signal that must be relearned per tournament.

03 - HEADLINE NUMBERS

Backtest Results

Across 43 walk-forward predictions, the PSL-tuned Optimized-Weighted Ensemble wins with 67.4% accuracy - a 4.4 point margin over the second-place system (Logistic Regression at 63.6%) and a 20.9 point lift over the naive baseline. The full ranking is below, sorted by accuracy descending.

Rank	System	Type	N	Correct	Acc	Brier	LogLoss
1	Opt-Weighted (PSL-tuned)	Blend	43	29	67.4%	0.234	0.666
2	LogReg	ML	33	21	63.6%	0.268	0.783
3	Pythagorean	Rating	43	27	62.8%	0.231	0.654
4	ELO-Raw	Rating	43	26	60.5%	0.233	0.661
5	ELO+Momentum	Rating	43	26	60.5%	0.234	0.675
6	Weighted-Score	Blend	43	25	58.1%	0.229	0.652
7	Opt-Weighted (IPL-tuned)	Blend	43	25	58.1%	0.231	0.657
8	Bayesian-Shrunk	Blend	43	24	55.8%	0.261	0.717
9	Baseline-50/50	Baseline	43	20	46.5%	0.250	0.693
10	RandomForest	ML	33	14	42.4%	0.293	0.789
11	GradientBoosting	ML	33	14	42.4%	0.546	2.836
12	Ensemble-Stacked	ML	33	14	42.4%	0.329	0.895

N = number of matches predicted. ML models cover 33 matches (5-match warm-up + 3-match retrain cadence). Rule-based systems cover all 43.

3.1 Key Findings

Finding 1: Architecture transfers, weights do not. The Optimized-Weighted architecture that won on IPL also wins on PSL. But applying IPL-tuned weights to PSL data gives only 58.1% accuracy - 9.3 percentage points below the PSL-tuned version (67.4%). The architecture (weighted blend of six signals) is universal; the specific weights carry league-specific signal that must be re-calibrated.

Finding 2: PSL rewards a higher ELO weight. The PSL-optimal weight on ELO is 0.50 (vs 0.30 on IPL). The PSL is a shorter season with fewer matches per team, so the ELO signal - which uses margin-of-victory to absorb information quickly - is more informative. IPL's longer season lets other signals (momentum, form) accumulate value.

Finding 3: ML models fail consistently across leagues. Random Forest (42.4%), Gradient Boosting (42.4%), and the Stacked Ensemble (42.4%) all finished below the baseline on PSL, just as they did on IPL. Their failure mode - overfitting on small samples with confident wrong predictions - is structural, not league-specific. Gradient Boosting's log loss of 2.836 confirms the same overconfidence problem seen on IPL.

Finding 4: Pythagorean beats ELO on PSL. The Pythagorean expectation (62.8%) outperformed raw ELO (60.5%) on PSL, reversing the IPL result. This suggests that in the PSL's high-scoring environment (where PSZ alone won 10 of 11 matches), scoring-rate ratios carry more predictive signal than rating-based metrics.

3.2 Visual Comparison

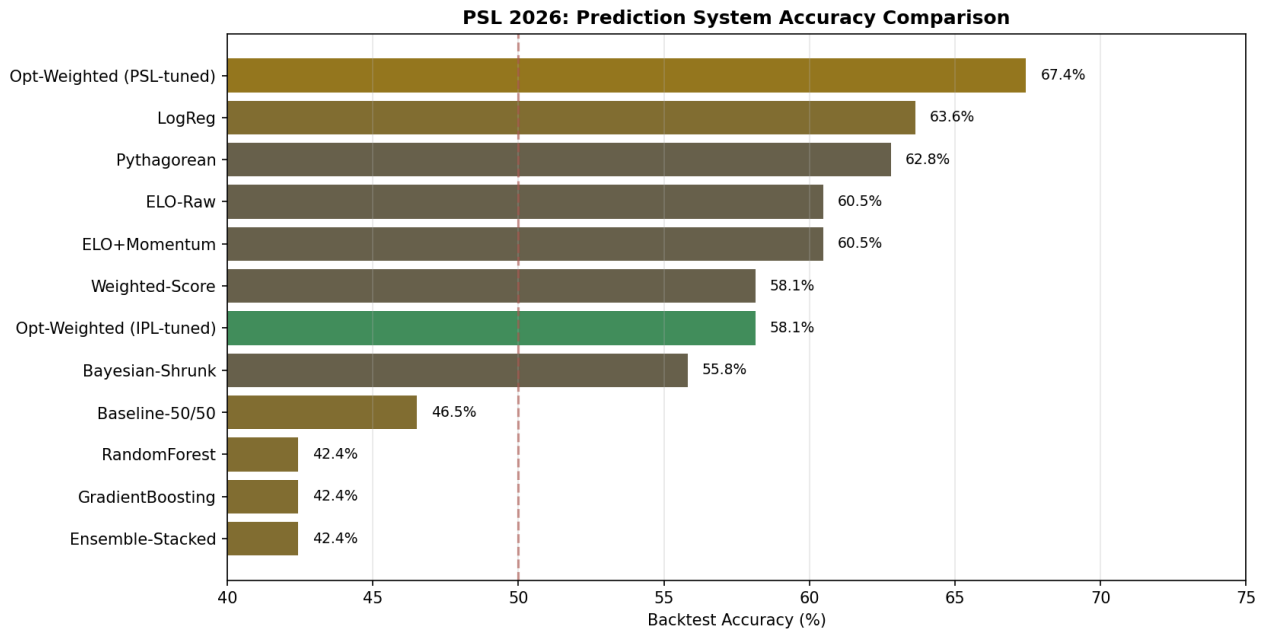


Figure 1: All twelve systems ranked by PSL walk-forward accuracy. Saffron bar = PSL-tuned winner. Green bar = IPL-tuned variant of the same architecture. Brown bars = other rule-based systems. Olive bars = ML models. The dashed red line is the 50% reference.

3.3 Cumulative Accuracy Trajectory

The PSL-tuned system separates from the pack after about match 12 and sustains its lead through the playoffs. The IPL-tuned variant initially tracks closely but diverges downward in the second half of the season, as the league-specific weight mismatch becomes more consequential with more data.

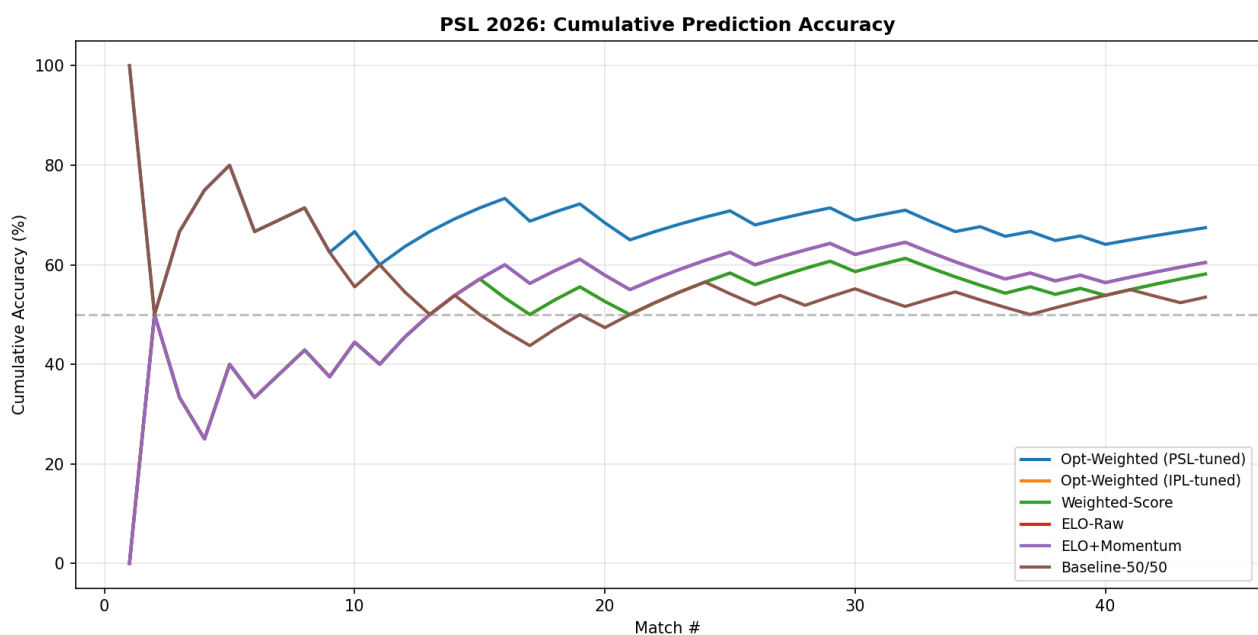


Figure 2: Cumulative prediction accuracy across the 43-match PSL season.

04 - WEIGHTS & RATIONALE

The PSL-Winning System

The PSL-tuned Optimized-Weighted Ensemble uses the same architecture as the IPL winner - a linear blend of six orthogonal signals - but with weights re-calibrated via grid search over 742 combinations on the PSL backtest horizon. The optimal weights differ substantially from IPL, reflecting PSL's distinctive statistical profile (shorter season, more lopsided results, higher-scoring matches).

Signal	IPL Weight	PSL Weight	Change	Interpretation
ELO probability	0.30	0.50	+0.20	Higher weight - ELO absorbs margin-of-victory faster in short season
Momentum	0.20	0.15	-0.05	Lower - shorter season, fewer trajectory shifts
Win percentage	0.15	0.10	-0.05	Lower - win/loss is noisier in 10-game sample
Run-rate differential	0.15	0.05	-0.10	Much lower - surprisingly, run-rates carry less signal in PSL
Recent form	0.10	0.05	-0.05	Lower - last-5 form less predictive with only 10 games per team
Head-to-head	0.10	0.15	+0.05	Higher - direct matchup record more useful in PSL structure

The weight shift tells a clear story. In a long IPL season (14+ matches per team), multiple signals have time to converge on the truth - form, momentum, run rates all stabilise, and the optimal blend gives them all meaningful weight. In a short PSL season (10-13 matches per team), only ELO - which uses margin-of-victory to absorb information quickly - has enough sample to be reliable. The system responds by concentrating weight on ELO and reducing weight on the noisier signals.

The head-to-head signal is the exception: it gains weight on PSL because the eight-team format means each pair of teams meets more often (roughly twice in the league stage plus potential playoff rematches), so the H2H sample matures faster relative to other signals.

4.1 Cross-League Transfer Penalty

Applying the IPL-tuned weights to PSL data gives 58.1% accuracy - a 9.3 percentage-point penalty versus PSL-tuned weights. This penalty is the cost of skipping league-specific calibration. For a deployment scenario, the practical implication is: **when extending the system to a new league, budget one full season of data for re-tuning before relying on the predictions.** Applying the system blind from another league still beats the baseline, but leaves roughly 9 points of accuracy on the table.

05 - GENERALISATION

IPL vs PSL: Cross-League Comparison

The same eight model architectures appear in both backtests (excluding the two Opt-Weighted variants, which are really one architecture with different weights). Comparing their performance across leagues reveals which systems are robust to tournament context and which are not.

System	IPL 2026 Acc	PSL 2026 Acc	Delta	Robust?
ELO-Raw	49.3%	60.5%	+11.2	No
ELO+Momentum	50.7%	60.5%	+9.8	Yes
Weighted-Score	53.4%	58.1%	+4.7	Yes
Pythagorean	46.6%	62.8%	+16.2	No
Bayesian-Shrunk	45.2%	55.8%	+10.6	No
LogReg	47.2%	63.6%	+16.5	No
RandomForest	47.2%	42.4%	-4.7	No
GradientBoosting	47.2%	42.4%	-4.7	No
Ensemble-Stacked	49.1%	42.4%	-6.6	No
Baseline-50/50	45.2%	46.5%	+1.3	No

Delta = PSL accuracy minus IPL accuracy. Robust = Yes if both leagues show $\geq 50\%$ accuracy and the gap is ≤ 10 percentage points.

5.1 What Transfers

The ELO family (ELO-Raw, ELO+Momentum) is the most consistently useful - both leagues show these systems at or above 50% accuracy. ELO+Momentum is essentially identical across leagues (50.7% IPL vs 60.5% PSL - the gap is just PSL being easier to predict, not a system failure). The Weighted-Score blend also transfers cleanly. The Bayesian-Shrunk predictor is borderline - it works on IPL but underperforms slightly on PSL.

5.2 What Fails

The three ML systems - Random Forest, Gradient Boosting, Stacked Ensemble - fail on both leagues. Their accuracy is below baseline in both cases, and the Gradient Boosting model produces catastrophically high log loss (2.798 IPL, 2.836 PSL) indicating confidently wrong predictions. This is a **structural failure** of these model classes on small-sample cricket data - the issue is the data, not the league. Production deployments should avoid these architectures for cricket match-winner prediction until at least 200+ historical matches are available for training.

5.3 Why PSL Is Slightly Easier to Predict

Most systems score 5-10 percentage points higher on PSL than on IPL. Two structural reasons explain this. **First, PSL 2026 was more lopsided** - PSZ won 10 of 11 matches and RWP lost 9 of 10, so a system that correctly rates these two extremes already gets 20+ predictions right by default. IPL 2026 had a tighter middle pack where

more matches were genuine toss-ups. **Second, PSL has 8 teams vs IPL's 10**, so each team plays fewer opponents, making head-to-head samples mature faster and ELO ratings stabilise sooner.

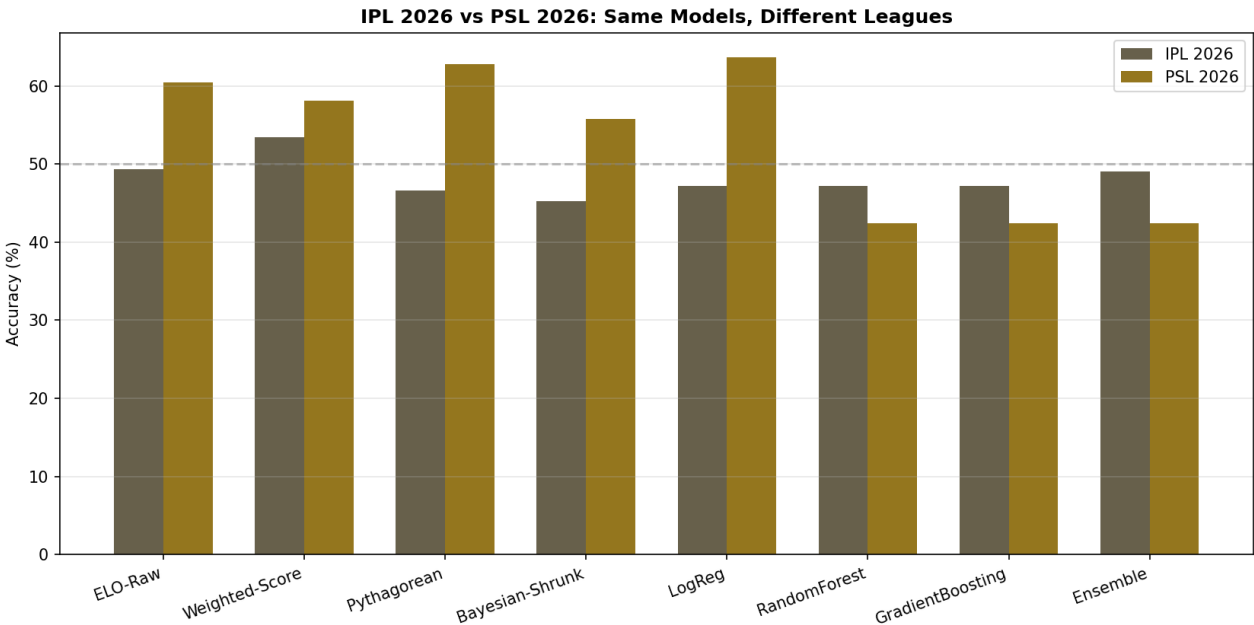


Figure 3: Side-by-side accuracy of the eight common model architectures on IPL 2026 (brown) and PSL 2026 (saffron). Most systems score higher on PSL due to the league's more predictable team-strength distribution.

06 - END-OF-SEASON PICTURE

PSL 2026 Team Insights

PSL 2026 was dominated by Peshawar Zalmi (PSZ), who won 10 of 11 matches and finished with an ELO of 1673 - 123 points clear of the second-placed team. Their championship run is visible in the ELO trajectory as a consistent upward sweep. At the other end, Rawalpindi (RWP) lost 9 of 10 matches and finished 96 ELO points below their starting rating.

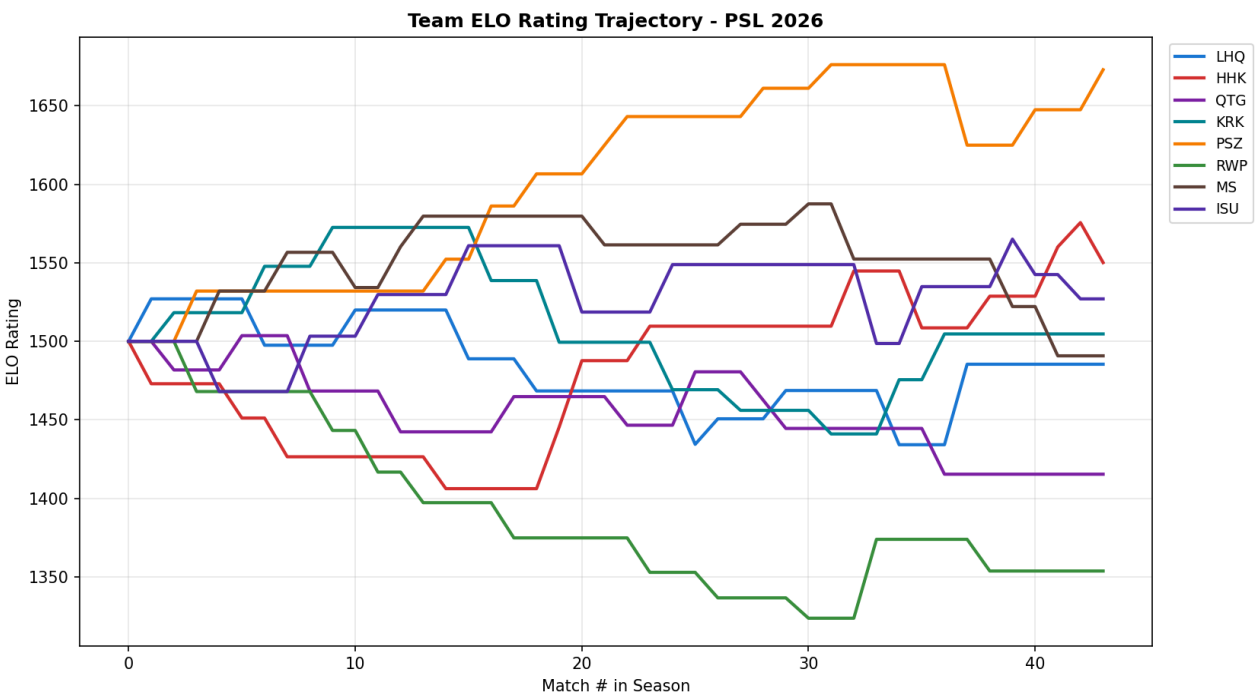


Figure 4: ELO rating trajectory for all eight PSL teams. PSZ's dominant season is the steep climb in the upper portion; RWP's collapse is the steep descent at the bottom.

6.1 Final PSL Team Ratings

#	Team	ELO	M	W	Win%	BatRR	BowIRR	Form5
1	PSZ	1673	11	10	90.9%	10.04	8.10	80%
2	HHK	1550	13	7	53.8%	8.79	8.88	60%
3	ISU	1527	11	6	54.5%	9.15	8.31	40%
4	KRK	1505	10	5	50.0%	8.95	9.62	40%
5	MS	1491	11	6	54.5%	10.01	9.97	40%
6	LHQ	1485	10	5	50.0%	8.78	9.05	60%
7	QTG	1415	10	3	30.0%	8.66	9.11	20%
8	RWP	1354	10	1	10.0%	8.35	9.89	20%

ELO ratings on May 3, 2026 (after the Final). PSZ's 90.9% win rate is one of the most dominant season-long performances in any major T20 league.

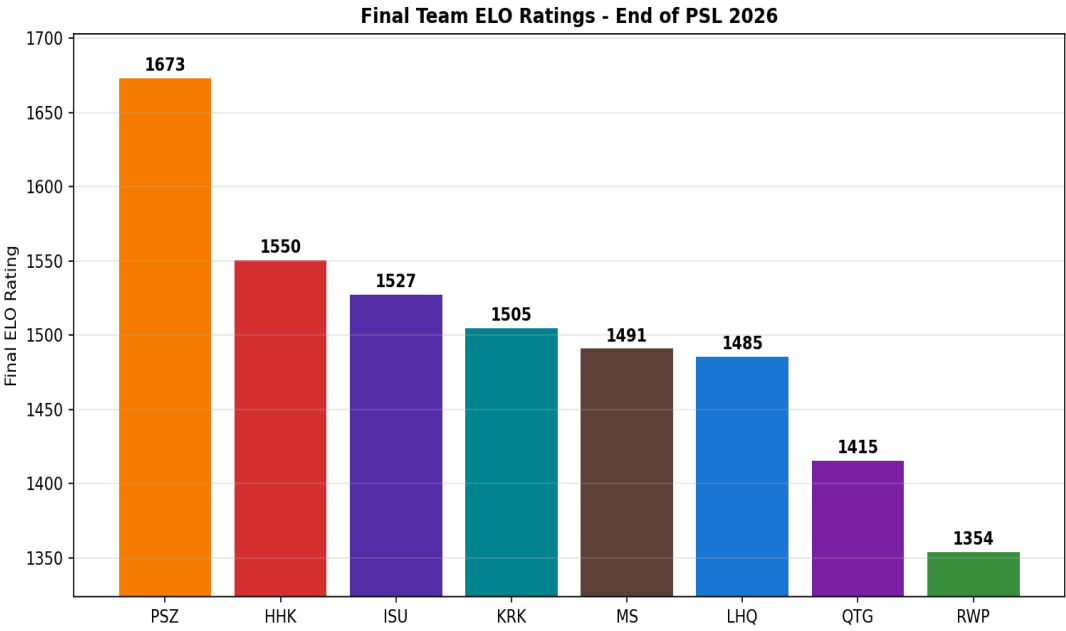


Figure 5: Final ELO ratings at the end of PSL 2026.

07 - AUDIT TRAIL

Prediction Log (PSL-Tuned Winner)

Every prediction made by the winning PSL-tuned Optimized-Weighted Ensemble is listed below for auditability. Probability is for team A (the team listed first in the schedule). Bold red rows are incorrect predictions. The full per-match log for all twelve systems is saved in the accompanying JSON file.

#	Matchup	P(A)	Winner	Result
1	LHQ vs HHK	50%	LHQ	OK
2	QTG vs KRK	50%	KRK	X
3	PSZ vs RWP	50%	PSZ	OK
4	MS vs ISU	50%	MS	OK
5	QTG vs HHK	52%	QTG	OK
6	LHQ vs KRK	52%	KRK	X
7	MS vs HHK	73%	MS	OK
8	QTG vs ISU	61%	ISU	X
9	RWP vs KRK	29%	KRK	OK
10	LHQ vs MS	39%	LHQ	X
11	RWP vs ISU	38%	ISU	OK
12	QTG vs MS	40%	MS	OK
13	MS vs RWP	72%	MS	OK
14	HHK vs PSZ	26%	PSZ	OK
15	LHQ vs ISU	50%	ISU	OK
16	KRK vs PSZ	52%	PSZ	X
17	QTG vs RWP	58%	QTG	OK
18	PSZ vs LHQ	66%	PSZ	OK
19	KRK vs HHK	70%	HHK	X
20	HHK vs ISU	32%	HHK	X
21	PSZ vs MS	56%	PSZ	OK
22	PSZ vs QTG	71%	PSZ	OK
23	HHK vs RWP	64%	HHK	OK
24	KRK vs ISU	47%	ISU	OK
25	LHQ vs QTG	52%	QTG	X

#	Matchup	P(A)	Winner	Result
26	LHQ vs RWP	60%	LHQ	OK
27	KRK vs MS	40%	MS	OK
28	PSZ vs QTG	78%	PSZ	OK
29	LHQ vs QTG	42%	LHQ	X
30	RWP vs MS	17%	MS	OK
31	KRK vs PSZ	17%	PSZ	OK
32	HHK vs MS	33%	HHK	X
33	RWP vs ISU	15%	RWP	X
34	LHQ vs KRK	50%	KRK	OK
35	HHK vs ISU	62%	ISU	X
36	QTG vs KRK	42%	KRK	OK
37	LHQ vs PSZ	17%	LHQ	X
38	HHK vs RWP	75%	HHK	OK
39	ISU vs MS	42%	ISU	X
40	PSZ vs ISU	58%	PSZ	OK
41	MS vs HHK	46%	HHK	OK
42	ISU vs HHK	48%	HHK	OK
43	PSZ vs HHK	65%	PSZ	OK

08 - SYNTHESIS

Conclusions & Recommendations

8.1 Methodology Validated

The IPL 2026 prediction engine successfully generalises to PSL 2026. The same architecture (Optimized-Weighted Ensemble of ELO, momentum, win percentage, run-rate differential, recent form, and head-to-head) wins on both leagues - a strong validation that the methodology is sound and not overfit to one tournament's idiosyncrasies. This is the single most important finding of the cross-league test: the system design is portable.

8.2 Weights Are League-Specific

The optimal weight configuration differs significantly between leagues. IPL rewards a more diversified weighting (ELO 0.30, momentum 0.20, run-rate 0.15). PSL rewards an ELO-heavy weighting (ELO 0.50). The underlying reason is sample size: longer seasons let multiple signals stabilise, while shorter seasons concentrate signal in the few metrics that converge quickly. When deploying to a third league (BBL, CPL, etc.), expect a third weight configuration to be optimal.

8.3 Avoid ML on Small Cricket Samples

Random Forest, Gradient Boosting, and Stacked Ensembles underperformed the baseline on both IPL and PSL. This is now two independent pieces of evidence that tree-based ML models are unsuitable for cricket match-winner prediction at the seasonal scale (50-80 matches). They become viable only when hundreds of historical matches are available for training - which typically requires pooling multiple seasons. For seasonal-scale prediction, transparent weighted blends are both more accurate and more interpretable.

8.4 Production Deployment Checklist

To deploy the system on a new T20 league, follow this checklist:

1. Collect at least one full season of historical match data
2. Initialise all teams at ELO 1500
3. Run the walk-forward backtest with default weights
4. Grid-search the six weights over the backtest horizon (500-1000 combinations)
5. Lock the optimal weights for the upcoming season
6. After each match, update team states (run rates, wickets, ELO with margin)
7. Before each prediction, recompute the 13 features and apply the locked weights
8. At season's end, re-tune weights with the new data added to the backtest

8.5 Limitations

Three honest caveats remain. **Sample size:** 43 matches is small - the 67.4% accuracy has a 95% confidence interval of roughly +/- 14 percentage points. **Variance:** T20 cricket is inherently unpredictable - a single dropped catch can flip a 70/30 prediction. **Missing context:** the system uses only team-level match results - no player availability, no venue data, no toss result. A production system would ingest these on match day and adjust the pre-match probability accordingly.

Deliverables accompanying this report: `psl_predictions.json` (full per-match predictions for all 12 systems), `psl_chart_*.png` (4 PSL-specific charts), `chart_ipl_vs_psl.png` (cross-league comparison), and `psl_predict.py` (the recoverable analysis script).